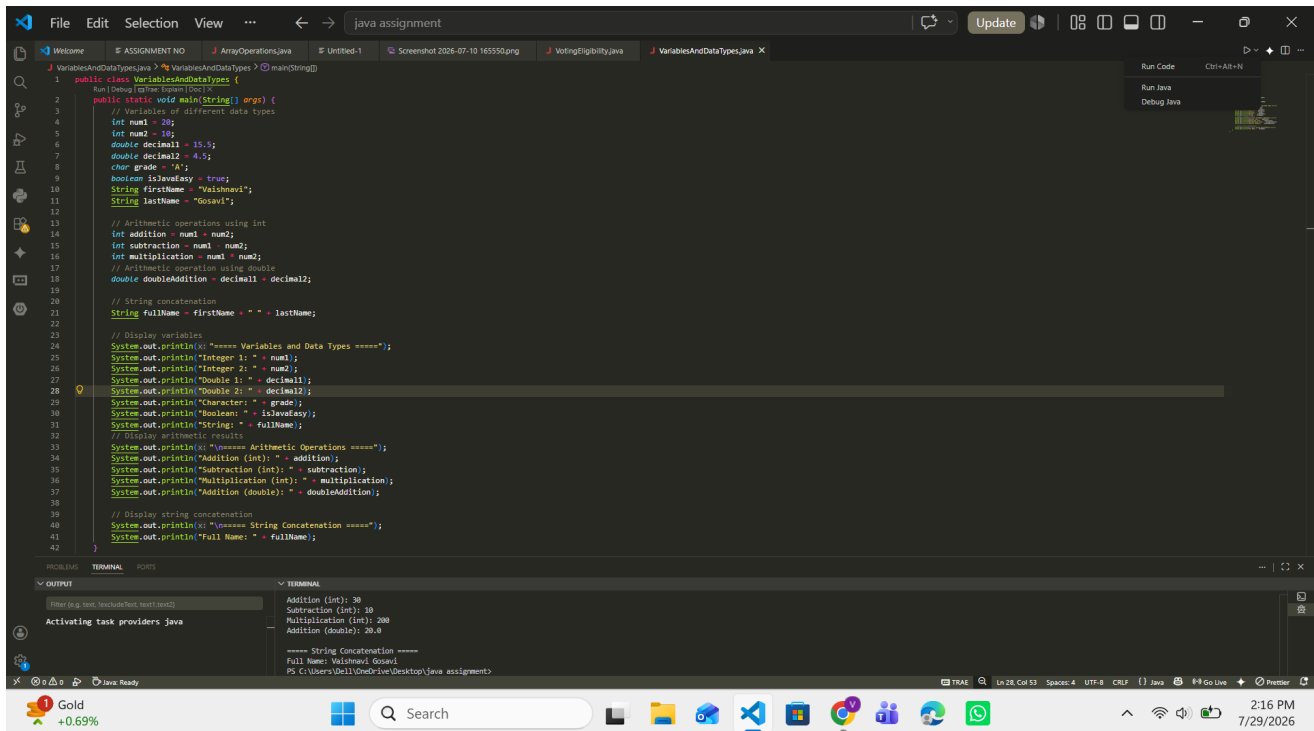


Java Programming Assignment 1

Question 1 - Variables and Data Types



```
1 public class VariablesAndDataTypes {
2     public static void main(String[] args) {
3         // Variables of different data types
4         int num1 = 20;
5         int num2 = 10;
6         double decimal1 = 15.5;
7         double decimal2 = 4.5;
8         char grade = 'A';
9         boolean isJavaEasy = true;
10        String firstName = "Vaishnavi";
11        String lastName = "Gosavi";
12
13        // Arithmetic operations using int
14        int addition = num1 + num2;
15        int subtraction = num1 - num2;
16        int multiplication = num1 * num2;
17        // Arithmetic operation using double
18        double doubleAddition = decimal1 + decimal2;
19
20        // String concatenation
21        String fullName = firstName + " " + lastName;
22
23        // Display variables
24        System.out.println("==== Variables and Data Types =====");
25        System.out.println("Integer 1: " + num1);
26        System.out.println("Integer 2: " + num2);
27        System.out.println("Double 1: " + decimal1);
28        System.out.println("Double 2: " + decimal2);
29        System.out.println("Character: " + grade);
30        System.out.println("Boolean: " + isJavaEasy);
31        System.out.println("String: " + fullName);
32
33        // Display arithmetic results
34        System.out.println("==== Arithmetic Operations =====");
35        System.out.println("Addition (int): " + addition);
36        System.out.println("Subtraction (int): " + subtraction);
37        System.out.println("Multiplication (int): " + multiplication);
38        System.out.println("Addition (double): " + doubleAddition);
39
40        // Display string concatenation
41        System.out.println("==== String Concatenation =====");
42        System.out.println("Full Name: " + fullName);
43    }
44 }
```

Terminal Output:

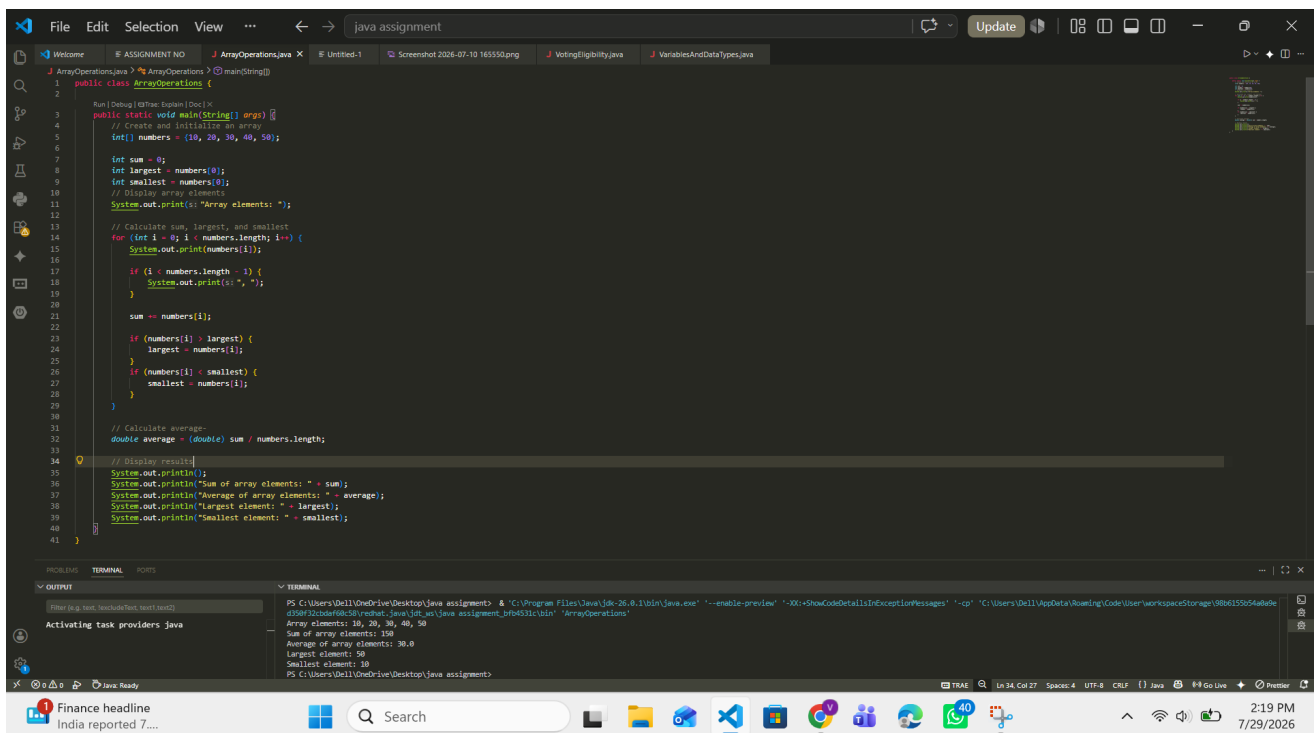
```

Addition (int): 30
Subtraction (int): 10
Multiplication (int): 200
Addition (double): 20.0

==== String Concatenation =====
Full Name: Vaishnavi Gosavi

```

Question 2 - Array Operations



```
1 public class ArrayOperations {
2     public static void main(String[] args) {
3         // Create and initialize an array
4         int[] numbers = {10, 20, 30, 40, 50};
5
6         int sum = 0;
7         int largest = numbers[0];
8         int smallest = numbers[0];
9
10        // Display array elements
11        System.out.println("Array elements: ");
12
13        // Calculate sum, largest, and smallest
14        for (int i = 0; i < numbers.length; i++) {
15            System.out.println(numbers[i]);
16
17            if (i < numbers.length - 1) {
18                System.out.print(", ");
19            }
20
21            sum += numbers[i];
22
23            if (numbers[i] > largest) {
24                largest = numbers[i];
25            }
26            if (numbers[i] < smallest) {
27                smallest = numbers[i];
28            }
29        }
30
31        // Calculate average
32        double average = (double) sum / numbers.length;
33
34        // Display results
35        System.out.println();
36        System.out.println("Sum of array elements: " + sum);
37        System.out.println("Average of array elements: " + average);
38        System.out.println("Largest element: " + largest);
39        System.out.println("Smallest element: " + smallest);
40    }
41 }
```

Terminal Output:

```

PS C:\Users\Devi\OneDrive\Desktop\Java assignment> java -cp . ArrayOperations
Array elements: 10, 20, 30, 40, 50
Sum of array elements: 150
Average of array elements: 30.0
Largest element: 50
Smallest element: 10

```

Question 3 - Voting Eligibility

